

LIQUID-BASED DYNAMIC COUNTERWEIGHT SYSTEM TO OPTIMIZE ELEVATOR ENERGY USAGE BASED ON LOAD.

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FACULTY OF ENGINEERING

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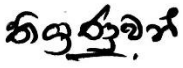
AUGUST 2025

DECLARATION OF AUTHORSHIP

I, M.K.T Gunawardena declare that this dissertation titled “LIQUID-BASED DYNAMIC COUNTERWEIGHT SYSTEM TO OPTIMIZE ELEVATOR ENERGY USAGE BASED ON LOAD”. and the content presented in it, is my own.

I confirm that,

- This work was done wholly or mainly in candidature for a degree at this University.
- Where I have consulted the published work of others, which has always been clearly attributed.
- Where I have quoted from the work of the others, the source is always given.
- Except for such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.

D/ENG/23/0075/MC M.K.T. Gunawardena 

Date: 12/10/2025

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ABSTRACT

Elevator systems are very important in modern buildings with many floors, but they use a lot of energy, which is a problem for sustainability. In normal elevators, the counterweight is fixed and designed to balance the weight of the cabin plus about 50% of the maximum load. However, this setup wastes energy when the number of passengers or load changes because the counterweight does not adjust to the real weight.

This project introduces a new idea — a liquid-based dynamic counterweight system that can automatically change its weight to match the cabin load in real time. The system uses load cells (with HX711 amplifiers) to measure weights, a NEMA 17 stepper motor for moving the elevator, and solenoid valves with a liquid transfer system to add or remove liquid from the counterweight tank.

A 1:7 scaled prototype was built in SolidWorks and tested as a three-story elevator model. An Arduino control unit reads the weight data and controls how much liquid moves between tanks to keep balance between the cabin and counterweight.

The system's performance was compared with a normal elevator by checking the motor's energy use. Results showed that the dynamic system reduced the motor's load, saved energy, and made the elevator move more smoothly and stably.

This study proves that using a liquid counterweight that adjusts automatically can make elevators more energy-efficient and eco-friendly.

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INTRODUCTION

Elevator systems are very important in tall buildings because they help people and goods move easily between floors. However, they use a lot of electricity, which makes them less environmentally friendly. Most elevators use a fixed counterweight to balance the cabin and part of its load. This helps reduce the motor's effort, but it is not efficient when the number of passengers or the weight in the cabin changes. If the load is heavier or lighter than expected, the motor has to work harder, which increases energy use and causes more wear on the system.

This project aims to solve that problem by creating a liquid-based dynamic counterweight system that can automatically change its weight depending on the cabin load. The system transfers liquid between tanks to adjust the counterweight, keeping the elevator balanced and moving smoothly.

The setup includes load cells with HX711 modules to measure the weight, a NEMA 17 stepper motor to move the elevator, and solenoid valves to control the flow of liquid. An Arduino microcontroller manages everything — it reads the sensor data and controls the motor and valves in real time. A 1:7 scale prototype of a three-story elevator was designed and built using SolidWorks to test the concept.

The performance was checked by comparing how much current the motor used with and without the dynamic counterweight. Results showed that the new system reduces motor effort, saves energy, and makes the elevator run more smoothly.

Overall, this project helps move toward smarter and more energy-efficient elevators by combining mechanical design with automatic control technology.

LITERATURE REVIEW

Elevator systems are essential in tall buildings, helping people and goods move quickly between floors. However, they use a large amount of energy, which is a growing problem in designing environmentally friendly buildings. Traditional elevators use a fixed counterweight to balance the cabin and part of its load. Although this helps reduce the motor's effort, it becomes inefficient when the cabin's actual load is much higher or lower than expected. In those cases, the motor must work harder, leading to more energy use, faster wear of mechanical parts, and lower efficiency.

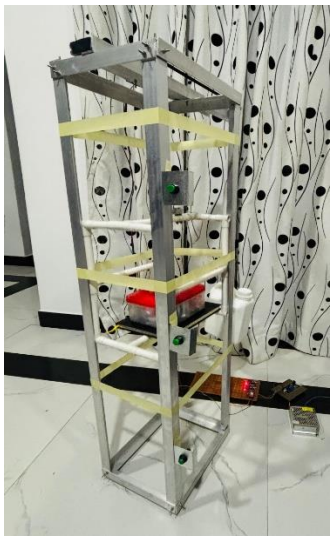
To make elevators more energy-efficient, engineers have tried different methods, such as regenerative braking (which recovers energy when the elevator goes down), variable frequency drives (which optimize motor operation), and smart control systems that adjust performance based on feedback. Some research has also explored adjustable counterweights and load sensors that detect the cabin's real-time weight to balance the system more effectively.

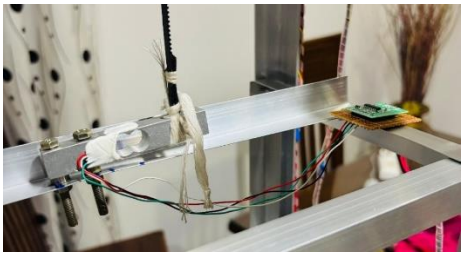
Mechanical systems like balance mechanisms and tension regulators have helped improve stability and reduce motor strain. However, these still rely on solid counterweights, which must be moved back into place, meaning they save little energy overall.

Only a few studies have looked into fluid-based counterweight systems, which can change weight more smoothly and accurately. A liquid-based system can automatically adjust its mass based on the cabin's load, improving balance, saving energy, and providing smoother operation.

This project focuses on developing such a liquid-based dynamic counterweight system. It uses load sensors to measure the cabin's weight and a liquid transfer mechanism to change the counterweight in real time. The goal is to reduce motor load, save energy, and support sustainable elevator designs that fit modern green building standards.

METHODOLOGY AND DESIGN

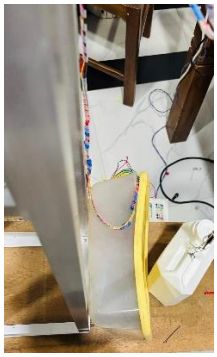





PROJECT OVERVIEW

This project focuses on developing a liquid-based dynamic counterweight system for elevators to improve energy efficiency and mechanical performance. Traditional elevator systems use fixed counterweights, which are effective only for average load conditions. When the cabin load changes, the motor must work harder, wasting energy and increasing wear. The proposed system uses load sensors, solenoid valves, and a liquid transfer mechanism to automatically adjust the counterweight mass in real time based on the cabin load. A 1:7 scaled prototype, designed in SolidWorks and controlled by an Arduino, was built to compare energy usage between static and dynamic counterweight systems.



- Why Choose This Project
    - Energy Efficiency: Reduces power consumption by optimizing the counterweight balance according to actual load conditions.
    - Innovation: Introduces a new fluid-based counterweight concept, rarely explored in elevator design.
    - Sustainability: Contributes to green building technologies and sustainable engineering practices.
    - Practical Application: Addresses real-world problems in elevator energy waste and motor wear.
    - Automation & Control: Integrates sensors and Arduino-based automation for smart load adjustment.
    - Research Potential: Opens possibilities for future studies in adaptive balancing and intelligent elevator systems.
    - Scalability: Can be developed further for industrial-scale elevators or other lifting mechanisms.

- Advantages of the Project
    - Improved Energy Efficiency: Automatically adjusts counterweight mass to match the cabin load, reducing motor effort and power consumption.
    - Dynamic Load Balancing: Maintains system balance in real time using a liquid transfer mechanism, ensuring smoother and more stable elevator motion.
    - Reduced Mechanical Stress: Minimizes strain on the motor, pulleys, and cables, extending the overall lifespan of components.
    - Supports Sustainability: Contributes to green building design by optimizing energy usage and promoting eco-friendly elevator technology.
    - Automation and Smart Control: Incorporates load sensors and Arduino-based automation for intelligent operation without manual intervention.
    - Smoother and Quieter Operation: Reduces vibrations and jerky movements often caused by unbalanced elevator loads.
    - Cost-Effective Solution: Uses simple components such as load cells, valves, and tanks — offering an affordable method for improving efficiency.
    - Research and Innovation Potential: Provides a new approach to adaptive balancing systems, opening opportunities for further development and integration with modern building management systems.

PROBLEM IDENTIFICATION AND SOLUTION DEVELOPMENT

In conventional elevator systems, counterweights are designed to balance the cabin weight plus approximately half of the rated load. While this approach improves energy efficiency compared to unbalanced systems, it remains inefficient when the actual cabin load differs from this fixed value. When fewer passengers or lighter loads are present, the counterweight becomes heavier than required, forcing the motor to consume extra power to move the cabin. Conversely, when the cabin is heavily loaded, the counterweight provides insufficient balance, again increasing the load on the motor.

This imbalance leads to higher energy consumption, increased wear and tear on mechanical parts, and reduced system efficiency. In addition, static counterweight systems cannot adapt to changing load conditions in real time. As energy efficiency and sustainability become major priorities in modern building design, there is a strong need for a more adaptive, energy-optimized counterweight mechanism that can automatically respond to varying elevator loads.

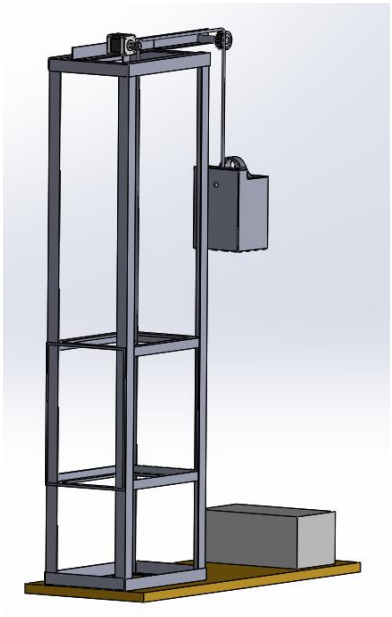
To overcome these challenges, this project introduces a liquid-based dynamic counterweight system that automatically adjusts its mass based on the real-time cabin load. The system uses load cells to measure the cabin and counterweight weights, and a liquid transfer mechanism—controlled by solenoid valves and an Arduino microcontroller—to shift liquid between tanks to maintain balance.

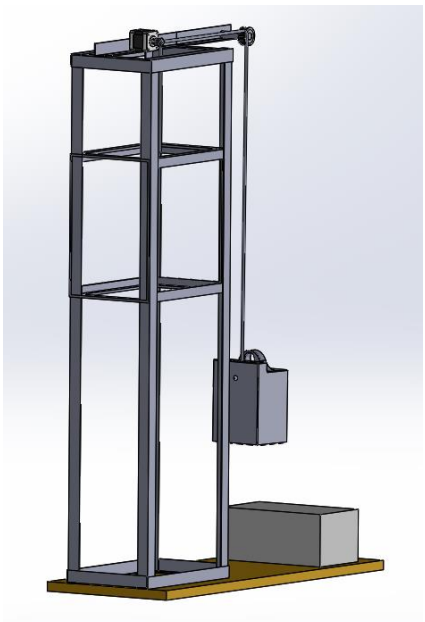
3D MODELING AND SIMULATIONS

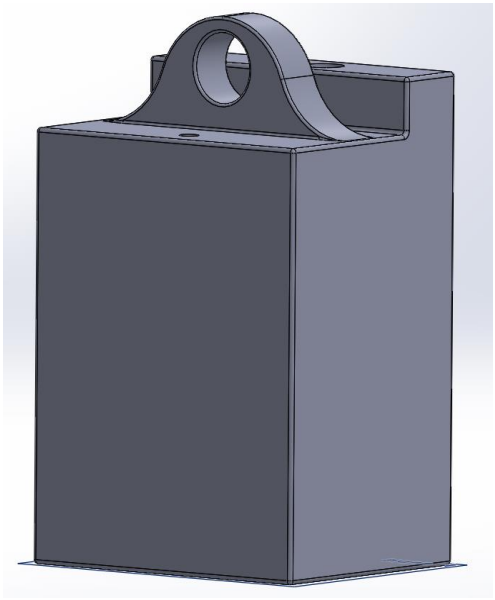
The liquid-based dynamic counterweight elevator system was completely designed and modeled using SolidWorks 3D CAD software. This stage was very important because it allowed the entire structure to be visualized before building it, helping plan the mechanical and structural layout accurately. The 3D model clearly showed how different parts—such as the elevator cabin, frame, counterweight tank, pulleys, stepper motor, and liquid transfer system—work together during operation.

Each component, including the elevator platform, tank system, support frame, motor mount, and pulley assembly, was first modeled separately and then assembled to create the full system. In SolidWorks, the positions of the shafts, alignment of the motor, and belt paths were carefully adjusted to ensure smooth movement and proper balance between the cabin and counterweight. The assembly tools in the software helped identify and fix possible issues like part collisions, misalignments, or space limitations before making the physical prototype.

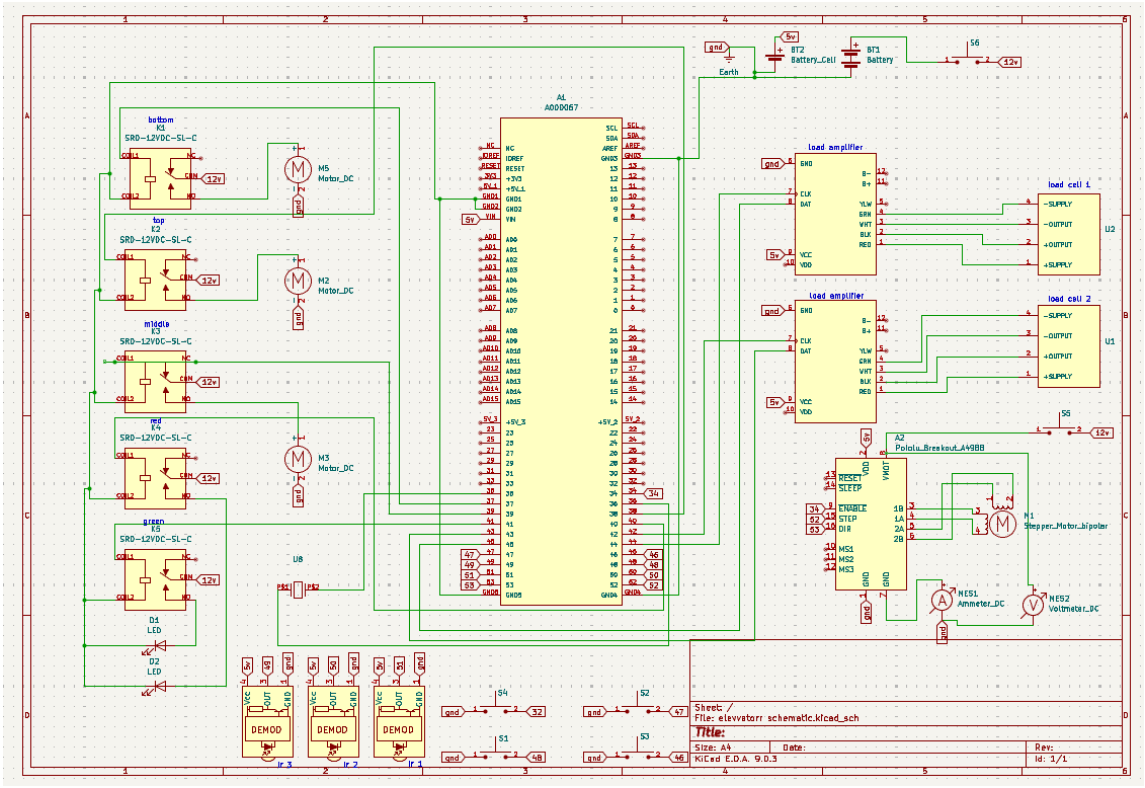
Motion simulations were also carried out to visualize the elevator's vertical movement and how the liquid shifts between tanks for balancing. These simulations helped study how the system responds to different cabin loads and confirmed that the design would work mechanically. SolidWorks features such as mass property analysis and center of gravity checks were used to optimize the design, ensuring that the system remained stable and safe.







SCHEMATIC WIRING DIAGRAM



MATERIAL SELECTION AND ANALYSIS.

Materials used,

- Frame – Aluminum
- Base – wood
- Enclosure – Polymate box
- Counter tank - Plastic box
- Pipelines – Hydraulic hose

The materials for the liquid-based dynamic counterweight elevator system were selected based on strength, rigidity, weight, cost, availability, and ease of construction. The goal was to make the system strong, stable, and safe to operate while keeping it affordable and easy to build as a small-scale prototype.

At first, PVC tubes were used for the elevator cabin frame because they are light and easy to assemble. However, during testing, the PVC frame bent slightly under load, causing unstable movement. To fix this problem, the frame was rebuilt using aluminum L-bars, which are strong, rigid, and lightweight. Aluminum also resists bending, is easy to machine, and gives a neat, durable finish—making it ideal for the prototype.

The main support structure was made using mild steel box bars, chosen for their high strength, low cost, and good weldability. Mild steel provided a stable base for the elevator's moving parts and counterweight system, ensuring the structure could handle both static and moving loads.

Mounting brackets for the stepper motor, pulleys, and sensors were made from sheet metal and aluminum plates because they are light, easy to drill, and suitable for fastening components.

The counterweight tank and liquid reservoir were 3D printed using PLA plastic, which is light, smooth, and accurate for making detailed parts. Flexible PVC hoses were used to transfer liquid between tanks, ensuring smooth flow and preventing leaks.

The pulleys and 3M timing belt were selected to match the NEMA 17 stepper motor, providing precise movement with low vibration and minimal slippage. For sensors, infrared (IR) and ultrasonic sensors were used to detect the elevator's position and liquid levels because they are accurate, reliable, and easy to connect with the Arduino controller.

CALCULATIONS

CALCULATIONS BASED ON STATIC COUNTER WEIGHTS

Maximum Elevator Mass = 5kg

Counter weight mass = 2.25kg (Approximately 45%)

Pulley Radius = 0.03

Forces due to Weights = (Melevator−Mcw) * G= (5 – 2.5) * 9.81 = 2.5*9.81 = 24.525 N

Frictional forces = 0.10 * (5 * 9.81)= 4.905N (considering 10% of weight)

Net force = 24.525 + 4.905 = 29.43N

Torque required = R * Fnet = 0.03 * 29.43 = 0.8829Nm

CALCULATIONS BASED ON DYNAMIC COUNTER WEIGHTS

Maximum Elevator Mass = 5kg

Counter weight mass = 5kg

Pulley Radius = 0.03

Maximum stepper motor torque = 0.45Nm (Nema 17 1.7A)

Forces due to Weights = (Melevator−Mcw) * G = 0 N

Friction= 0.1 * Melevator * G = 0.1 * 5 * 9.81 = 4.905N

Net force = 4.905N

Torque required = R * Fnet = 0.03 * 4.905 = 0.147115Nm

Conclusion;

When the counterweight mass is equal to the elevator car's mass, the required torque to lift the system is significantly reduced compared to the case with static counterweights.

SCALING CALCULATIONS

Elevator car platform

:- Structural frame Scale height = = 0.122 =

:- Elevator car Scale height = = 0.143 =

:- Elevator car Scale width = = 0.27 =

:- Elevator car Scale length = = 0.143 =

The prototype structure, the elevator platform dimensions of 0.3×0.3×0.2 cm, is designed as an approximate 1:7 scale model of a typical 3-story elevator system. Assuming an average floor height of 3 m and real cabin dimensions of 2.1×1.1×1.4 m, the proportions of the model closely replicate those of a real-world elevator,

MOTOR PUMP CALCULATIONS

Pump flow rate = 120 L/h = 0.033L/s

Density of water = 1000kg/m3

Ex:- Weight of the load = 500g

Time to fill up the counter weight tank

Time = Volume / Flow rate = 0.5 / 0.0333

= 15s approximately

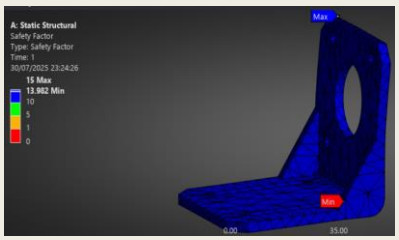
I have implemented a stop time for the motor to calculate the weights in between,

Final time to fill 500g = 18s approximately.

FEA SIMULATIONS.

FINITE ELEMENT ANALYSIS SIMULATIONS

- Finite element analysis of stepper motor bracket



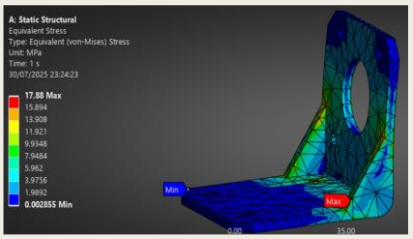
SAFETY FACTOR

MAX = 15
MIN = 13.982



TOTAL DEFORMATION

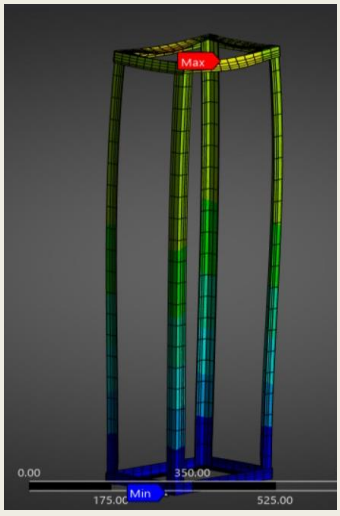
MAX = 0.0263mm
MIN = 0 mm



EQUIVALENT STRESS

MAX = 17.88MPa
MIN = 0.0029MPa

- Finite element analysis of elevator frame



TOTAL DEFORMATION

MAX = 0.0000591mm
MIN = ---



SAFETY FACTOR

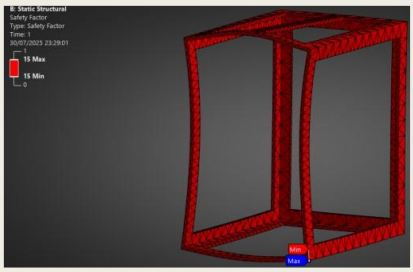
MAX = 15
MIN = 15



EQUIVALENT STRESS

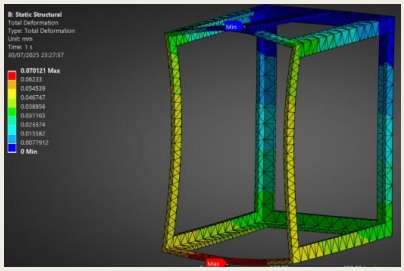
MAX = 0.32168 MPa
MIN = 2.6113E-5 MPa

➢ Finite element analysis of elevator car platform



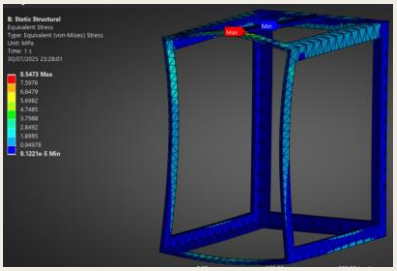
SAFETY FACTOR

MAX = 15
MIN = 15



TOTAL DEFORMATION

MAX = 0.0701 mm
MIN = 0 mm



EQUIVALENT STRESS

MAX = 8.5473 MPa
MIN = 9.122e-5 MPa

Conclusion;

Based on the FEA analysis of the structural component including safety factor, total deformation, and equivalent stress it can be concluded that the system operates within a safe region.

TESTING AND VALIDATIONS

Motor Current Measurement

Static counter load measurements





Dynamic counter load measurements





Steady state load measurements



Conclusion,

The liquid-based dynamic counterweight system reduced current draw compared to the static design, proving that the project objectives of enhancing efficiency were successfully achieved.

SWITCHING BETWEEN MODES

Static Counterweight Mode

Works as a traditional elevator with fixed counterweights.

Counterweight remains unchanged regardless of cabin load.

Higher motor current draw observed.



Dynamic Counterweight Mode

Activates dynamic liquid based counterweight adjustment.

Load sensors detect cabin weight in real time.

Liquid transfer mechanism adjusts counterweight mass automatically.



DISCUSSION AND RESULTS

The liquid-based dynamic counterweight elevator system was created to solve the energy waste and imbalance problems found in normal elevators that use fixed counterweights. In regular elevators, the counterweight is designed to balance the cabin plus about 50% of its rated load. When the real cabin load changes, the motor has to work harder to lift or lower the car, which increases energy use and causes more wear. The goal of this project was to design, simulate, and build a small working model that can automatically change its counterweight using liquid, reducing motor strain and saving energy.

The project started with a detailed 3D design made in SolidWorks. This was very helpful in understanding how all the parts—such as the elevator, counterweight, tanks, and pulleys—fit and work together. Each part, including the elevator cabin, motor mount, pulley, base frame, counterweight tank, and water pipes, was modeled separately and then assembled into the full system. The 3D simulation helped check motion paths, clearances, and possible collisions before building. It also allowed the team to study stress on the frame, confirming that the aluminum structure could safely handle a 5 kg load with little bending.

At first, PVC tubes were used for the elevator cabin because they were light and cheap. But tests showed that PVC was too weak, bending and shaking during movement. To fix this, the frame was rebuilt with aluminum L-bars, which were stronger and more stable while still being light. This made the elevator move more smoothly and accurately. The final version was strong, light, and suitable for repeated testing.

The liquid transfer system was a key innovation. It used two tanks — one for the counterweight and another at the top of the frame — with solenoid valves and relays to control water movement. HX711 load cells measured the weight of both the elevator and the counterweight. These sensors were connected to an Arduino Mega, which read the data and opened or closed valves to move water between tanks until the system was balanced. This real-time adjustment kept the elevator stable even when the cabin load changed.

During early tests, there were leaks in the liquid system because of poor sealing in the 3D-printed parts and flexible pipes. This problem was solved by using threaded connectors with rubber gaskets and silicone sealant, which made the system leak-proof. The improved setup allowed smooth and consistent liquid movement between tanks.

The control system, built on Arduino, was programmed to manage both static and dynamic modes. In static mode, the elevator worked like a normal system with a fixed counterweight. In dynamic mode, the Arduino constantly read sensor data and automatically adjusted the counterweight. The program was designed with different states such as load detection, balancing, movement, and idle, to ensure smooth transitions. IR sensors were placed at different levels to detect the elevator's position, while ultrasonic sensors monitored water levels to avoid overflow or running dry. This made the system fully automatic and safe to use.

Testing involved measuring the current drawn by the NEMA 17 stepper motor in both static and dynamic modes. Results showed that the dynamic counterweight system reduced the torque needed by the motor, meaning it used less energy and stayed cooler during operation. The current was measured using an ammeter connected in series with the motor's power supply for accurate results.

There were some technical problems during testing, such as vibration affecting the weight readings, IR sensors not working well under bright light, and delays in liquid transfer. These issues were fixed one by one: the vibration problem was solved by mounting load cells on a rigid base with rubber dampers, the sensor issue was fixed with protective covers and signal filtering in the code, and the liquid transfer delay was reduced by adjusting valve timing and recalibrating flow rate.

Scaling was another key factor. The prototype was built at about 1:7 scale of a real three-story elevator. This size was chosen so tests could be done in a lab while still keeping the correct proportions for movement and load. The scale was calculated based on height, cabin size, and counterweight dimensions to make the model's behavior realistic.

Efficiency tests showed that the dynamic counterweight design used noticeably less current compared to the static system. This proved that automatically adjusting the counterweight helps save energy. The sensors and Arduino also made control more accurate and repeatable, reducing human error and setup time.

The project used locally available materials and affordable components, such as the stepper motor, solenoid valves, and sensors, keeping costs low while maintaining strength and reliability.

Despite its low cost, the system worked effectively and consistently, proving that such designs can be used in small buildings or educational projects.

In conclusion, the project successfully met its goals of designing, building, and testing a liquid-based dynamic counterweight elevator. It worked well in both static and dynamic modes, showing clear energy savings and smoother operation. The mix of mechanical balance, sensor feedback, and automated control made it safe and efficient. The study also suggests future improvements like better sensor calibration, refined liquid flow control, and IoT-based monitoring for tracking energy use in real time. Overall, this project lays the groundwork for developing smarter and more energy-efficient elevator systems that reduce power use and mechanical stress.

CONCLUSION

This project successfully designed and developed a liquid-based dynamic counterweight elevator system aimed at optimizing energy usage through automatic mass balancing. The main objective was to reduce the elevator's energy consumption by adjusting the counterweight mass in real time according to the cabin load. A 1:7 scaled prototype was designed using SolidWorks, fabricated with aluminum L-bars, and controlled via an Arduino Mega microcontroller.

The system utilized HX711 load sensors to detect cabin and counterweight mass and a liquid transfer mechanism to dynamically maintain balance. A NEMA 17 stepper motor powered the cabin movement, while IR and ultrasonic sensors provided accurate position and liquid-level feedback. Experimental testing showed the prototype performed effectively under both static and dynamic modes, demonstrating noticeable improvements in motor performance and reduced energy consumption during dynamic balancing.

Initially, the elevator cabin was constructed using PVC pipes, but due to vibration and instability, it was upgraded to an aluminum frame for better rigidity and durability. Water leakage issues were resolved through improved sealing using silicone and gaskets. Comparative tests confirmed that the liquid-based counterweight system significantly reduced motor current draw compared to a static counterweight, validating the system's energy efficiency.

The Arduino-based control logic successfully synchronized all system components—sensors, motor, and solenoid valves—ensuring smooth automatic balancing during load variations. Overall, the project demonstrated that a fluid-based counterweight system can effectively enhance elevator efficiency compared to traditional mechanical designs.

Key objectives achieved:

1. Design and simulate a scaled elevator with a liquid-based dynamic counterweight system.
2. Implement automatic mass balancing using load sensors and a fluid transfer mechanism.
3. Evaluate energy efficiency by comparing static and dynamic counterweight performance.

The final prototype proved to be structurally stable, functionally reliable, and energy-efficient, presenting a promising approach toward sustainable elevator systems for modern buildings.

RECOMMENDATIONS

1. Improved Liquid Flow Control

Use precision flow control valves or peristaltic pumps for smoother, faster, and more accurate fluid transfer during balancing.

2. Integration of Regenerative Braking

Incorporate regenerative braking to recover energy during descent, enhancing overall system efficiency and supporting green building standards.

3. Wireless Monitoring and Data Logging

Upgrade to an ESP32 or similar Wi-Fi-enabled controller to allow real-time data visualization, remote monitoring, and performance tracking via a dashboard or mobile app.

4. Advanced Sensor Calibration and Filtering

Apply Kalman filtering or moving average algorithms to reduce noise in load cell readings and improve real-time accuracy.

5. Safety Enhancements

Add limit switches, emergency stop circuits, and thermal protection for motors and relays to ensure system safety and reliability.

6. Full-Scale Model Development

Develop a full-scale prototype to validate the system under real-world conditions, including full cabin load, travel distance, and speed.

7. Material Optimization and Aesthetics

Utilize lightweight, corrosion-resistant materials such as aluminum composites or stainless steel to improve durability and visual appeal.

8. Integration with Smart Building Systems

Incorporate the elevator into IoT-based building management systems for automatic energy optimization based on occupancy and load demand.

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APPENDICES

BUDGET

• Structural Support Frame	= 4800
• Elevator Platform	= 970
• NEMA 17 Stepper Motor	= 3000
• Stepper Motor Bracket	= 360
• Stepper Motor Driver	= 280
• Pulley Mechanism	= 680
• Controlled Solenoid Valve	= 1400
• Storage Water Tank	= 1600
• Infrared (IR) Sensors	= 600
• Ultrasonic Distance Sensor	= 300
• Load sensor with amplifier module	= 1500
• 3M Timing Belt	= 1500
• Microcontroller Unit	= 4400
• Electrical Enclosure	= 700
• Regulated Power Supply Unit	= 1400
• Other Fabrication Expenses	= 3000
• Wooden base	= 1200
• Counter weight tank	= 600
• Hydraulic pipes	= 450
• Water sealant	= 600
• Transport	= 5500
• Switches	= 300
• Hydraulic pumps	= 1860
• Ammeter / Voltmeter	= 920
• Electrical wires	= 1200
• Wire covers	= 800
• Relay modules	= 1450
• Buck convertor	= 600
TOTAL	= 41970